

MISTIREMARIAM GIRMA

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PROFESSIONAL SUMMARY

Mechanical Engineering student at Addis Ababa University with international academic, research, and industrial experience spanning robotics, aerospace maintenance, automation, manufacturing, and sustainable engineering. Demonstrated technical acumen through research at Institut Pascal (France) and internships at Ethiopian Airlines Group and Abugida Robotics. Proficient in SolidWorks, ROS2, MoveIt, NVIDIA Isaac Sim, and Python, with a strong interest in advancing aerospace systems, robotic manipulation, and circular manufacturing.

EDUCATION

Sigma Clermont	<i>Clermont-Ferrand, France</i>
<i>Exchange Student – Mechanical Engineering (Robotics Lab Research)</i>	2026
Addis Ababa University	<i>Addis Ababa, Ethiopia</i>
<i>Bachelor of Science in Mechanical Engineering</i>	Expected August 2027

ENGINEERING & RESEARCH EXPERIENCE

Institut Pascal Robotics Laboratory – Sigma Clermont	<i>Clermont-Ferrand, France</i>
<i>Engineering Research Intern</i>	2026
- Developed robotic workflows for an autonomous waste-sorting collaborative robot (cobot) to optimize automated segregation. - Designed robotic disassembly strategies for attached waste components to support circular manufacturing and material recovery. - Simulated robotic manipulation using NVIDIA Isaac Sim and CoppeliaSim; implemented motion planning via ROS2 and MoveIt. - Assisted PhD research on deformable object manipulation through automated shoe sole disassembly.	
Ethiopian Airlines Group – Directorate of Component Maintenance	<i>Addis Ababa, Ethiopia</i>
<i>Engineering Intern</i>	2025
- Rotated through multiple aircraft component maintenance workshops, gaining core exposure to aerospace MRO operations. - Assisted in component inspection, repair, and overhaul while strictly adhering to international aviation quality and safety standards. - Participated in reverse engineering initiatives to support local tool/part redesign and minimize reliance on import replacement components.	
Abugida Robotics	<i>Addis Ababa, Ethiopia</i>
<i>Engineering Intern</i>	2024
- Contributed to automation projects integrating Arduino, Raspberry Pi, various physical sensors, and embedded systems. - Assisted in rapid prototype development, system integration, technical troubleshooting, and performance verification testing. - Completed Machine Learning training and built a functional supervised learning classification model using Python.	
Zelaki Project (Arizona State University & University of Gondar)	<i>Ethiopia</i>
<i>Mechanical Engineering Team Member</i>	2023
Supported the fabrication, mechanical assembly, troubleshooting, and maintenance of community-scale plastic recycling machinery.	

TECHNICAL PROJECTS & RESEARCH

- Friction Measurement Device:** Modeled the entire mechanical assembly in SolidWorks, fabricated the physical prototype, and integrated an Arduino-based automation and data collection control system.
- Automated Ventilation System:** Engineered a prototype for automatic thermal regulation and modeled heat transfer performance dynamically using MATLAB script simulations.
- Automation in Mining Research:** Conducted an investigative study on deploying industrial automation and robotic safety technologies to eliminate human risk and improve operational efficiency in high-hazard mining environments.

TECHNICAL SKILLS

Robotics & CAD:	ROS2, MoveIt, NVIDIA Isaac Sim, CoppeliaSim, SolidWorks, AutoCAD
Programming & HW:	Python, MATLAB, Arduino, Raspberry Pi, Embedded Systems, Machine Learning
Mechanical & Mfg:	Machine Design, Reverse Engineering, HVAC, Fluid Mechanics, Assembly, Prototype Fabrication
Languages:	Amharic (Native), English (Professional), French (Beginner A1)

LEADERSHIP, AWARDS & CERTIFICATIONS

- **Leadership:** Managing Director of Book Club • Ethiopian Space Science Society (ESSS); Executive Member • MiNT Club.
- **Honors & Programs:** CNN Academy Graduate; Yale Young African Scholars (YYAS); UNECA/ITU Coding Program; UNECA Innovation Contest.
- **Certifications:** Power Learn Project Graduate; HVAC System Design (ESME); ML & Deep Learning; Cybersecurity (ADBI).
- **Presentations:** 20th ESSS General Assembly (Lectured on Hydrogen-powered aircraft, Electric aircraft, and AI in dark matter).